



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/34

Paper 3 Further Mechanics

May/June 2025

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.



- 1 A light spring of natural length a and modulus of elasticity $20mg$ is placed so that it stands vertically on a horizontal plane. The lower end of the spring is fixed to the plane. A particle of mass m is attached to the upper end of the spring.

The particle is pushed vertically downwards until the length of the spring is $\frac{3}{5}a$. The system is then released from rest.

Find the maximum extension of the spring in the subsequent motion. [5]

[illegible]

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- 2 A particle P of mass m kg moves along a horizontal straight line against a resistive force of magnitude $2mv^3$ N, where v ms⁻¹ is the velocity of P at time t s. When $t = 0$, $v = 1$.

(a) Find an expression for v in terms of t . [4]

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(b) Find the displacement of P from its initial position when $t = 6$. [3]

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- 3 A particle P is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at any subsequent time t are denoted by x and y respectively.

(a) Derive the equation of the trajectory of P in the form

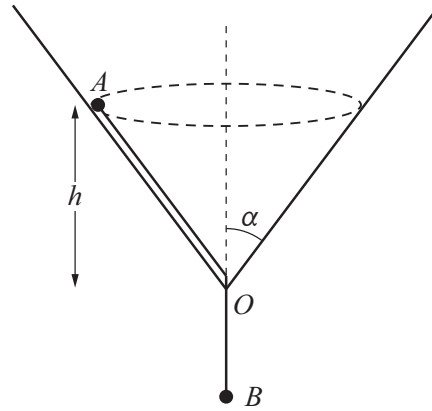
$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha. \quad [4]$$

[illegible]



This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

4



A hollow cone with a smooth inner surface is fixed with its vertex O downwards. The semi-vertical angle of the cone is α , where $\tan \alpha = \frac{3}{4}$. A light inextensible string has a particle A of mass m attached to one end and a particle B of mass m attached to the other end. The string passes through a small hole in the cone at O . Particle B hangs in equilibrium below O . Particle A is on the inner surface of the cone at a height h above the level of O and moves in horizontal circles with constant angular speed ω (see diagram).

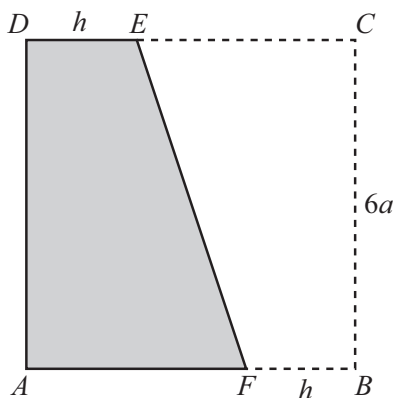
Find ω in terms of g and h .

[6]

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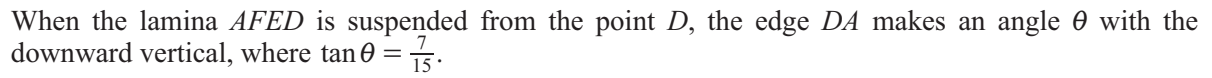
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$ABCD$ is a uniform square lamina of side $6a$. Points E and F are on DC and AB respectively and are such that $DE = FB = h$. The quadrilateral $BCEF$ is removed from the square lamina (see diagram).

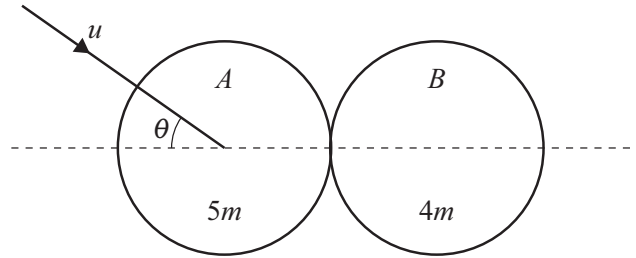
- (a) Show that the distance of the centre of mass of the resulting lamina $AFED$ from AD is $\frac{h^2 - 6ah + 36a^2}{18a}$ and find a corresponding expression for the distance of the centre of mass from AB . [5]

[illegible]



This image shows a full page of a document template designed for handwriting practice or note-taking. It consists of approximately 28 evenly spaced horizontal dotted lines across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

6



Two uniform smooth spheres A and B of equal radii have masses $5m$ and $4m$ respectively. Sphere A is moving with speed u on a horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e .

- (a) Show that the speed of B after the collision is $\frac{5}{9}u(1+e)\cos\theta$. [3]

[illegible]

After the collision the kinetic energy of A is equal to the kinetic energy of B .

(b) Given that $\tan \theta = \frac{2}{3}$, find the value of e .

[6]

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